# Use LLMs to Build Repair Tools, Not Patch Generation: Cost-Effective Automated Program Repair

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## Abstract

Large language models (LLMs) repair buggy submissions to programming assignments well, but they are deployed *one query per submission*: every buggy program triggers a fresh model call, so cost, latency, and external dependence scale with enrollment and recur every term, while the specification-level reasoning being purchased is identical across a class. Caching and cheaper models shrink only the constant, not the structure. We propose CoEP, which moves the LLM from the marginal-cost to the fixed-cost column: *spend the model once per assignment, not once per submission*. An LLM agent reads the assignment and a clustered sample of worked (buggy $\rightarrow$ fixed) pairs and authors the assignment-specific layer of a local repair tool: AST rewrite rules named after the misconceptions they fix, atop a shared symbolic substrate of generic edits and reference-guided minimal-hunk transplant from the class's own correct solutions. Every candidate patch is validated against the instructor's tests before it is kept; the tool can never self-report success, and submissions it abstains on escalate to a conventional per-submission call, so coverage is never sacrificed. On the standard benchmark of this literature, the synthesized tool repairs *more* submissions with *smaller* patches than the strongest hand-built symbolic tool, on that tool's own corpus and metric, with the entire model spend collapsed into a one-off synthesis plus a small escalation tail, so everything repaired locally costs no model call at all, offline and deterministically. A price-invariant break-even analysis shows when amortization wins and why cheaper models and caching do not change the answer, and the artifact yields what a per-query API cannot: reproducible minimal patches and a per-class misconception profile.


## Introduction

Timely, individualized feedback is the scarcest resource in programming education: introductory courses enroll hundreds of students on campus and tens of thousands online, each producing buggy submissions no instructor can examine one by one. Automated program repair (APR) has therefore long been proposed as a feedback engine for programming assignments: given a buggy submission and the assignment's test suite, produce a small patch that makes it correct while preserving the student's own approach (Yi et al. 2017; Gulwani, Radicek, and Zuleger 2018; Wang, Singh, and Su 2018; Hu et al. 2019; Ahmed et al. 2022).

Three generations of systems have pursued this goal. Early systems established the setting: test-driven repair whose patch must stay close to the student's program, because an unrelated rewrite passes the tests but teaches nothing (Yi et al. 2017). A second generation mined the class's own solution space, matching each buggy program against previously accepted solutions (Gulwani, Radicek, and Zuleger 2018; Wang, Singh, and Su 2018; Hu et al. 2019; Ahmed et al. 2022). This repairs cheaply, but it requires a large solution pool and fails whenever a submission matches no reference (Orvalho, Janota, and Manquinho 2025). LLM-based repair removed both constraints and now fixes most real student programs at modest per-bug prices (Zhang et al. 2024, 2022; Xia and Zhang 2024; Liu et al. 2024).

What the third generation fixed in coverage, it broke in cost structure. The deployment default is one model query per buggy submission, so spending grows with enrollment and restarts every term. The queries themselves are massively redundant. A single assignment is attempted by hundreds of students whose failures cluster into a few recurring misconceptions (Gulwani, Radicek, and Zuleger 2018; Hu et al. 2019), and on the corpus we study, about two thirds of a repair prompt is byte-identical across submissions to the same assignment. Paying a model to rediscover the same off-by-one for the three-hundredth time is not intelligence at work; it is a billing artifact. The standard remedies do not change this shape. Cheaper models and provider caching shrink the constant (Anthropic 2025; OpenAI 2024), and retrieval enriches each query (Dai et al. 2026), but every submission still triggers a model round trip with its marginal cost, latency, and data egress. The question is not how much a call costs, but where in the pipeline the LLM is invoked.

Our answer is to spend the LLM once per assignment, not once per submission. Agent research has shown that LLMs can author reusable code and amortize it across many uses (Cai et al. 2024; Wang et al. 2024), and classical synthesis has learned per-assignment fix rules from examples (Rolim et al. 2017). CoEP (Cost-Effective Program repair) instantiates this idea for assignment repair. An LLM agent reads the assignment once and authors its problem-specific layer: AST rewrite rules named after the misconceptions they fix.

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These rules run on a shared symbolic substrate, a generic edit library plus reference-guided transplant from the class's correct pool. Every candidate patch must pass the instructor's tests before it is kept, so a tool can never self-report success. The frozen tool then repairs submissions locally in about a second at zero marginal LLM cost, and the few it abstains on escalate to a conventional per-submission call.

The resulting artifact beats the hand-built state of the art on its own benchmark and metric: 97.75% repair rate at 0.317 mean relative patch size, against Refactory's 90.8% at 0.40, over 1,686 real student programs. We are explicit about attribution. The rule layer produces about 13% of local fixes, at near-minimal patch size and with a misconception label each; the transplant engine produces the rest, and an ablation shows the symbolic substrate alone already carries the coverage. The claim is therefore not that an LLM wrote a state-of-the-art repairer. The claim is the invocation structure: one cheap build plus a small fallback tail replaces an inference per submission, and the artifact it feeds exceeds the hand-built tool.

The stakes scale with deployment. In one course the per-submission bill is dollars; on an online judge it is not. The public Codeforces dump our benchmark draws from aggregates 12.6M submissions to over 10,000 problems (Penedo et al. 2025), At the standard \$3/\$15 per Mtok rate of the models we run, our measured per-submission repair costs about \$125 per thousand submissions, re-paid as every new submission arrives. The comparison is also robust to pricing. If a build costs $C_{\text{build}}$ and locally resolves a fraction $(1-f)$ of submissions that would each have cost $\Delta$, amortization wins past $N^* = C_{\text{build}}/(\Delta(1-f))$, and $N^*$ is invariant to token price when build and query spending follow the same schedule. The artifact also offers what per-submission inference cannot at any price. Every patch is re-executed against the instructor's tests before it is emitted, so it can never be invalid or unverified code, and the rule layer edits by AST-located source-span splice rather than re-rendering the module. It is deterministic and runs offline, so student code never leaves the institution. Its rule-fired repairs aggregate into a per-class misconception profile that no API call returns.

This paper makes the following contributions:

- **A cost-structure reformulation of assignment repair.** We move the LLM from per-submission marginal cost to per-assignment fixed cost, with a price-invariant break-even analysis $N^* = C_{\text{build}}/(\Delta(1-f))$: amortization is a structural advantage, not a constant-factor one that cheaper models erase.
- **The CoEP system.** A builder agent synthesizes misconception-named AST rewrite rules through a generate–validate–refine loop (Madaan et al. 2023; Shinn et al. 2023), atop a shared substrate of generic edits and reference-guided minimal-hunk transplant. Every patch is independently test-validated, and abstentions escalate to a strategy-matched LLM fallback.
- **An empirical result on the baseline's own corpus and metric.** CoEP's local repair exceeds Refactory on both coverage and patch size at negligible one-off build cost, with an honest mechanism attribution, a framework-only ablation isolating the LLM layer's contribution, and per-builder analyses across two commodity models.
- **Artifacts.** We release the harness, dataset-import pipeline, prompts, synthesized repairers, and all raw model outputs.[1]

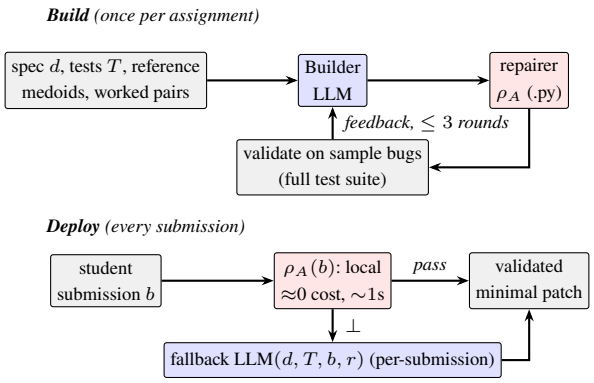



Figure 1: CoEP. LLM effort (blue) is spent once per assignment to synthesize a local repairer (red); at deployment, the large majority of submissions never touch a model.

## The Cost Anatomy of Per-Submission Repair

Decomposing the per-submission prompt of our PyDex reproduction shows where the money goes (Zhang et al. 2024; Xia and Zhang 2024). Averaged over the Refactory corpus, about two thirds of a query's input tokens are byte-identical across every submission to the same assignment: the instructions, the visible tests, and, in the zero-shot layout, the description. Only the buggy program and the retrieved exemplars vary. Provider caching cannot reach that redundancy, because it discounts a leading shared *prefix* rather than shared content. PyDex's layouts place the variable material first, so the longest identical prefix across submissions is about ten tokens, a few percent of the prompt. The remaining reuse is within a submission, across the $K$ sampled candidates, and even that is gated by cacheable minimums of 1,024 to 4,096 tokens that these 1,300-token prompts meet only at the smaller end.

Two consequences follow. First, caching is a discount, not a fix, and here not even a discount: the redundancy is real, but provider caching serves it at the wrong granularity. Second, the redundancy is exactly the amortizable part. What is constant per assignment, namely understanding the specification, the tests, and the space of likely mistakes, can be reified once into an executable artifact. That is the move CoEP makes, and the rest of the paper quantifies what it buys.

## Approach

### Problem Formulation

An *assignment* is a triple $A = (d, T, R)$: a natural-language description $d$, a test suite $T = \{(x_i, y_i)\}$ (function-entry or judge-style stdin/stdout), and the pool $R$ of correct student

[1] Anonymized repository submitted as supplementary material.

solutions, which the setting assumes is available at feedback time, as every per-assignment system from CLARA to Refactory does (Gulwani, Radicek, and Zuleger 2018; Hu et al. 2019). A *repair* of a buggy submission $b$ is a program $p$ that passes all of $T$. Patch quality is measured, never gated: RPS, the tree-edit distance between the ASTs of $b$ and $p$ normalized by the size of $b$, following Refactory's definition (Hu et al. 2019), is reported for every repair, since a small patch is feedback on the *student's* program while a rewrite passes tests but teaches nothing (Yi et al. 2017).

Conventional LLM-based APR maps each $b$ to a query $\mathrm{LLM}(d, T, b)$. CoEP instead synthesizes, once per assignment, a *repairer* $\rho_A : b \mapsto p \cup \{\bot\}$, an ordinary module with no model access, and deploys the hybrid: $\mathrm{repair}(b) = \rho_A(b)$ if $\rho_A(b) \neq \bot$ (local, $\approx$0 marginal cost), else $\mathrm{LLM}(d, T, b, r)$: the conventional call, strengthened with one reference $r$ chosen to match the student's strategy by clustering $R$ on program structure and drawing $r$ from the cluster nearest $b$.

## Repairer Synthesis (once per assignment)

**Build-set selection (local, no LLM cost).** The builder does not see the whole pool; it sees a *chosen* sample. We cluster the buggy pool on a structural view (TF-IDF over normalized-AST paths, identifiers folded) and, walking clusters largest-first, take from each the member with the smallest tree-edit distance to its nearest correct solution, a clean single-misconception exemplar rather than the medoid's typical but tangled program, paired with that nearest correct program. The result is a set of worked (buggy $\rightarrow$ fixed) pairs: frequent misconceptions first, each shown *with* its minimal fix, so a weak builder needs only to generalize a transformation it is shown, not infer it.

**Rule authoring.** The builder LLM receives $d, T$, reference solutions (structural-cluster medoids of $R$), and the worked pairs, and emits the assignment-specific rule module: AST rewrite rules, each named after the misconception it targets and tagged with the failure symptoms it addresses. The contract is strict. A rule locates its edit on the AST and splices it as a source-span replacement, never re-rendering the module, which would reformat untouched lines. It emits candidates rather than claimed fixes, and it never rewrites a whole function: a rule that cannot express a targeted edit must abstain so the submission escalates.

**Generate–validate–refine.** A single-shot module can contain its own defects, so the harness closes the loop: it loads the module, runs it on the sample bugs, *independently* validates every candidate patch by executing the full suite (the repairer's own claims are never trusted), and, while any sample remains unrepaired, returns the module with per-bug outcomes and the still-failing sources for revision, up to $R_{\max}$=3 builder calls. The best-validated round is kept, and build cost is exactly the token count of these exchanges, measured from the raw strings.

## Local Repair (every submission)

The frozen tool repairs in three stages, cheapest and smallest-patch first, validating every candidate against the full suite before it is kept:

**(1) Named rules + generic search.** Rules are *candidate generators*, composed by a symptom-ordered combination search: the submission's failing-test signature ranks rules whose declared symptoms match, then single rules, then pairs, under an explicit candidate budget. A shared generic library of comparison flips, off-by-one range and index shifts, and Boolean-operator flips is always on, so common slips are covered regardless of what the builder wrote; it is the inversion of the classical mutation-operator space (Just et al. 2014; Kim et al. 2013; Liu et al. 2019). Every patch that survives carries the ordered list of rule names that produced it, provenance that aggregates into a per-class *misconception profile*.

**(2) Reference-guided minimal-hunk transplant.** Most student bugs need a structural edit no token-level rule expresses; the correct pool supplies it. We diff $b$ against its structurally nearest correct solutions, rename the reference's variables to the student's (parameters by position, locals by first-appearance order), and adopt the *smallest* set of differing line-hunks that passes the tests, candidates ordered smallest-first and batch-validated, returning the minimum-RPS passer. This is reference reuse in the lineage of CLARA, Sarfgen, and Refactory (Gulwani, Radicek, and Zuleger 2018; Wang, Singh, and Su 2018; Hu et al. 2019), at hunk granularity rather than block granularity, which is what keeps patches smaller than the hand-built tool's.

**(3) Escalation.** A submission both stages abstain on takes the fallback $\mathrm{LLM}(d, T, b, r)$, its cost counted against CoEP in every comparison. Abstention is not the only trigger. Local repair's acceptance test is the instructor's suite, which says nothing about patch size, so transplant can return a patch larger than the student's whole program while the rule layer must abstain whenever it cannot express a targeted edit. We give transplant the same contract with a per-assignment threshold

$$\tau_p = \min\big(\mathrm{median}(\mathrm{RPS}_p) + 3\,\mathrm{MAD}(\mathrm{RPS}_p),\ 1\big), \quad (1)$$

computed once at build time from the patches the finished tool produces for that assignment's own buggy programs, counting only those that pass the tests. A patch above $\tau_p$ is an outlier for this assignment and is escalated; the smaller of the local and model patches is kept, so escalating can cost a call but never coverage or patch size. Calibration runs the tool and makes no model call, so the once-per-assignment claim is untouched, and $\tau_p$ is frozen with the tool rather than updated during deployment, which would make one submission's outcome depend on those before it. We use the median and MAD, not the mean and standard deviation, because the outliers being detected inflate the standard deviation enough to hide themselves, and we anchor at the median because MAD is a spread rather than a location. The multiplier 3 is the conventional outlier cutoff, not a fitted value; the firing rate is left to the data.

The cap exists because a purely relative threshold reads a bad distribution as normal. Where the correct pool is sparse the transplant is uniformly coarse, so the outlier rule sits high — 1.71 on cf_131A — and accepts patches larger than the program they repair as typical of the assignment. An RPS above 1 is a rewrite rather than a repair under any

assignment's standards, so the threshold never sits above it. The cap binds on five of the eleven assignments and never on the other six, where the outlier rule is already below it: it sleeps where the pool is dense and wakes where it is sparse.

**(4) Self-learning.** Each successful fallback fix can be generalized by the LLM into a new named rule, kept only if the augmented tool then repairs that miss on its own and rolled back otherwise. Absorption writes to a staging copy the evaluated pass never consults, so no submission's outcome depends on which submissions preceded it and both deployment configurations remain reproducible from the frozen artifact alone. The accumulated rules are a separate artifact, evaluated by a second frozen pass and priced to themselves rather than to the configuration they were gathered alongside (Section ). Self-learning spends model calls only on submissions that already escalated, so it too leaves the build claim untouched.

The local path runs on a CPU in $\sim$1–2 s per submission, deterministically and offline; a defective tool degrades to $\perp$ (escalation), never to a wrong "repair", because the grading harness re-validates every patch before a student sees it.

## Experiments

**Research Questions.** The primary research questions (RQs) addressed by our evaluation are as follows:

- **RQ1**: How effectively does CoEP repair buggy submissions compared to state-of-the-art symbolic and per-submission LLM approaches, in terms of repair rate and relative patch size?
- **RQ2**: To what extent does CoEP's once-per-assignment invocation structure reduce repair cost in turnaround time, tokens, and dollars, and past what submission volume does the one-off build pay off?
- **RQ3**: Can provider prompt caching and cheaper model tiers close the cost gap of per-submission repair, or does the gap persist because it is structural?
- **RQ4**: How does each component of CoEP, the LLM-authored rules and the shared symbolic substrate, contribute to its repairs and to the rules that fire in deployment?
- **RQ5**: How does the build-time escalation threshold $\tau_p$ affect the trade-off between patch size and model spend, and how sensitive is that trade-off to the outlier cutoff?
- **RQ6**: To what extent do the rules CoEP learns from its own escalations improve the frozen tool on submissions they were not induced from?
- **RQ7**: Does CoEP's advantage generalize across difficulty levels, from CS1 assignments to Codeforces problems rated 800 to 2000, and is it robust to the choice of LLM provider?

**Baselines.** **Refactory** (Hu et al. 2019) is the classical symbolic per-assignment tool: it refactors the class's correct solutions to match each buggy program's structure and transplants the divergent blocks. On its own corpus we use its published results; on Codeforces we run the released tool in its strongest configuration. **PyDex** (Zhang et al. 2024) is per-submission LLM repair: each buggy submission is sent to the model in its few-shot layout, which prepends the three most similar (buggy, correct) exemplars retrieved from the class pool, sampled candidates are filtered by the test oracle, and the surviving candidate closest to the student's program is returned. **PyDex$_{\mathbf{PC}}$** re-prices the same queries under provider prompt caching, as an upper bound that assumes a cache hit wherever the provider's rules allow one. **CoEP** is ours: a repairer synthesized once per assignment and model, evaluated in two deployment configurations plus one that measures what a term of use adds. **CoEP$_{\mathbf{L}}$** runs the frozen local tool alone, with zero LLM use at deployment, and accepts any patch its test oracle passes. That is deliberately the same acceptance rule Refactory uses, so the comparison between them is about repair, not about a size filter one of them applies. **CoEP$_{\mathbf{LF}}$** adds the per-submission fallback, which it spends both on abstentions and on patches above $\tau_p$ (Eq. 1); a gate is only meaningful where there is somewhere to escalate to, which is why it lives here rather than in the frozen configuration. **CoEP$_{\mathbf{L+SL}}$** is CoEP$_{\mathrm{L}}$ re-run with the rules self-learning accumulated, merged and frozen before evaluation so the arm is order-independent; the induction calls are priced to it rather than to CoEP$_{\mathrm{LF}}$, which never loads those rules. The submissions a rule was induced from are the only ones for which it is not a test of generalization, and are reported separately.

| Corpus | Problems | Buggy | Correct | Tests | LOC |
|---|---|---|---|---|---|
| Refactory | 5 | 1,783 | 2,442 | 7.4 | 11.9 |
| Codeforces | 6 | 1,187 | 698 | 35.5 | 12.3 |
| Total | 11 | 2,970 | 3,140 | 22.7 | 12.0 |

Table 1: Dataset statistics. Tests is the average test-suite size per problem; LOC is the average number of non-blank lines per submission.

**Datasets.** We evaluate on 11 problems and 2,970 buggy submissions drawn from two real-student corpora (Table 1). Refactory contributes five CS1-level assignments (Hu et al. 2019), the standard corpus of this literature (Ahmed et al. 2022; Zhang et al. 2024; Orvalho, Janota, and Manquinho 2025). Codeforces contributes six competitive-programming problems from the open-r1 release (Penedo et al. 2025), spanning difficulty ratings from 800 to 2000, which cover five official rating bands from Newbie to Candidate Master (Codeforces); we keep only problems whose complete official test suites are released, so the oracle is the judge's own. A label audit executes every submission against its full suite, and submissions labeled buggy that already pass are excluded from all repair targets, leaving 1,686 truly buggy Refactory programs. No train/test split is used, as prior per-assignment systems assume, and all results additionally hold with CoEP's build samples excluded from the denominator. On Codeforces every arm runs over the identical pools we built. On the Refactory corpus the arms we run share a pool, while Refactory itself is quoted from its paper and therefore carries its own denominator.

**LLM Configuration.** Every LLM arm runs on two models, one per provider: **GPT-5.4-mini** ($0.75/$4.50 per Mtok) and **Claude Sonnet 5** ($2/$10 per Mtok). We select each provider's cheapest tier at which prompt caching is functional, for two reasons. First, fairness to the caching baseline: $\text{PyDex}_{\text{PC}}$ only competes if caching can engage, and the absolute cheapest tiers rule it out. We verified empirically that Claude Haiku 4.5 refuses to cache prompts below its 4,096-token minimum, above these repair prompts, and that GPT-5.4-nano's automatic caching did not engage even on identical 1,200-token prompts, whereas both selected models cache at their documented 1,024-token minimums. Second, cheap tiers keep every measured dollar gap a conservative floor, and the break-even $N^*$ is price-invariant regardless (RQ3).

**Evaluation Metrics.** Metrics are grouped by the research question they serve.

**RQ1 and RQ7 (repair effectiveness). Repair rate (RR)**: percentage of buggy submissions for which the system produces a program passing the full test suite. **Relative patch size (RPS)**: tree-edit distance between buggy and repaired ASTs normalized by buggy program size, reimplemented from Refactory's definition (Hu et al. 2019); smaller patches preserve the student's approach and are pedagogically preferable. RPS is never used to *claim* a repair the tests do not support; it enters the pipeline only in $\text{CoEP}_{\text{LF}}$, where an outlying value routes a submission to the fallback (Eq. 1) and the smaller of the two patches is kept, so the gate can add a model call but cannot manufacture a repair or enlarge a patch. Repair rates carry 95% Wilson intervals. Comparisons between arms we ran ourselves are paired at the submission level and tested with exact McNemar, the exact form rather than the chi-square approximation because the discordant counts are often single-digit. That covers every pairing on Codeforces and every pairing against PyDex. It does not cover Refactory on its own corpus, where we quote its published per-assignment results and therefore have no per-submission outcomes to pair against; that one comparison is reported as unpaired rates with intervals, and we mark it as such wherever it appears.

**RQ2 and RQ3 (cost). Average turnaround time (ATT)**: mean wall-clock seconds to repair one submission end to end, decomposed into $\text{ATT}_{\text{cpu}}$ (local search on the CPU), $\text{ATT}_{\text{api}}$ (model-call latency), and their sum $\text{ATT}_{\text{all}}$; CoEP's one-off build wall-clock is reported separately as $\text{ATT}_{\text{build}}$ per assignment, an intercept like $C_{\text{build}}$ rather than a per-submission term. **Input/output tokens (InT/OuT)**: mean prompt and response tokens per repaired submission. **LLM cost**: USD spent on model inference per repaired submission and per assignment (including a system's one-off build cost), priced at the tiers above from exact prompt/response token counts. The name is deliberate: monetary inference spend is only one of the costs this paper argues about, and latency, determinism, and data egress are carried by ATT and the setup. The **abstention rate** $f$, the fraction of submissions the local tool escalates, is what the break-even $N^* = C_{\text{build}}/(\Delta(1-f))$ consumes. For RQ5 we additionally report the **gate rate**, the fraction of *repaired* submissions routed to the fallback by $\tau_p$, and sweep the outlier cutoff to show the resulting patch-size/spend frontier.

| Model | Method | RR (%) | RPS | ATT (s) | InT/OuT | LLM Cost ($ |
|---|---|---|---|---|---|---|
| *non-LLM* | Refactory | 85.92 | 0.500 | 5.63 | 0/0 | 0.00 |
| GPT-5.4 -mini | PyDex | 98.89 | 0.663 | 2.14 | 7,633/854 | 27.48 |
| | $\text{PyDex}_{\text{PC}}$ | 98.89 | 0.663 | 2.14 | 7,633/854 | 22.76 |
| | $\text{CoEP}_{\text{L}}$ | – | – | – | 0/0 | - |
| | $\text{CoEP}_{\text{LF}}$ | – | – | – | – | - |
| Claude Sonnet 5 | PyDex | 98.92 | 0.591 | 14.86 | 11,521/7,701 | 287.37 |
| | $\text{PyDex}_{\text{PC}}$ | 98.92 | 0.591 | 14.86 | 11,521/7,701 | 258.44 |
| | $\text{CoEP}_{\text{L}}$ | – | – | – | 0/0 | - |
| | $\text{CoEP}_{\text{LF}}$ | – | – | – | – | - |

Table 2: Main comparison on 2,872 buggy submissions (already-passing submissions excluded from every arm). RR and RPS are $N$-weighted micro-averages, RPS over repaired submissions; ATT is mean seconds per submission; InT/OuT are mean tokens per submission; cost is the total for the whole benchmark, and for CoEP it includes the one-off build. Refactory is model-independent, and its cell here is our run over both corpora; the 90.8%/0.40 we compare against in the text is its own published figure on its own corpus. $\text{CoEP}_{\text{L}}$ is the frozen local tool, $\text{CoEP}_{\text{LF}}$ adds the fallback on abstentions and on $\tau_p$-outlier patches; both are derived from a single evaluation pass. $\text{CoEP}_{\text{L+SL}}$ is reported separately (RQ6) because it is a second frozen pass rather than a rung of this one.

**RQ4 and RQ6 (component contributions).** The **repair-source split** attributes every fix to the mechanism that produced it (synthesized rule, generic rule, reference transplant, or LLM fallback). **Rule provenance** counts the rules the builder authored, the rules self-learning added, and the rules that actually fired at deployment, and tracks which recur across problems. For RQ6 the comparison is $\text{CoEP}_{\text{L}}$ against $\text{CoEP}_{\text{L+SL}}$ on identical submissions, reported with and without the few each rule was induced from. **Diagnosis coverage** is the fraction of repaired submissions whose patch carries a misconception label (rule-fired repairs), quantifying how much of the instructor-facing profile the tool actually delivers rather than leaving it implicit.

**Setup.** All experiments run on a single Apple M1 Ultra workstation (64 GB RAM, macOS) under Python 3.13, CPU only for everything but the API calls. PyDex uses $K{=}10$ sampled candidates per submission, issued in parallel, with every call billed and timed. The three CoEP configurations are derived from a single pass over the submission stream with no duplicated LLM calls, so the reported local numbers are always the frozen tool's. Refactory runs with its original algorithm unchanged, ported to Python 3. We release the code, prompts, raw model outputs, and per-submission logs.

## Results

We answer the five research questions stated in the Experimental Setup; together they cover the introduction's claims.

**RQ1: Effectiveness and Patch Quality** The central finding concerns CoEP's local repair against the strongest non-

LLM tool. On the Refactory corpus of 1,686 buggy submissions, CoEP's local path of named rules, the shared generic library, and reference-guided transplant repairs **97.75%** of submissions at mean **0.317** RPS, against Refactory's own reported **90.8%** at **0.40** on both axes: higher coverage and smaller patches, at zero marginal LLM cost. This comparison is unpaired, since Refactory's results on its own corpus are published per-assignment aggregates. Where we run Refactory ourselves, on Codeforces, the gap is wider and paired: 99.92% against 78.59%. This is the result that makes a local tool worth synthesizing. It is not merely cheaper than a symbolic tool that consumes every correct solution; it repairs more.

Against the per-submission LLM baselines the separation is not in coverage, which is near-saturated on CS1-scale programs (Zhang et al. 2024), but in patch size and cost. Over the whole benchmark $\text{CoEP}_{\text{LF}}$ reaches **99.55%** at mean RPS **0.461**, against PyDex's 98.89% at 0.591: it repairs more submissions and edits less of each, for \$4.15 against \$287.44 (Table 2). Because CoEP's patches are test-validated before a student sees them, its only local failure mode is abstention ($\perp$), never a wrong repair; abstentions take the per-submission LLM path, so the hybrid never sacrifices coverage.

Patch size is where the two corpora diverge sharply, and the direction is informative. On the CS1 corpus $\text{CoEP}_{\text{LF}}$ edits 0.289 of a program against PyDex's 0.358 and Refactory's 0.40. On Codeforces every system's patches inflate, and there Refactory's are the smallest (0.679 against 0.704 and PyDex's 0.920) — but it buys that by repairing 78.59% of submissions where the other two repair essentially all of them. A tool that declines the hard cases is measured only on the easy ones. We report both rather than choosing the framing that flatters us, and RQ7 decomposes it.

**RQ1.** On the Refactory corpus CoEP's local repair reaches 97.75% RR at 0.317 mean RPS, beating the classical non-LLM tool (90.8%, 0.40) on both coverage and patch size, exceeding its repair rate on every assignment, at zero marginal LLM cost, while the hybrid escalates abstentions so coverage is never lost.

**RQ2: Cost and Break-Even** The systems differ in *cost structure*, not just constant. A per-submission LLM arm (PyDex) pays an inference on every submission, so its cost grows linearly with enrollment and recurs every term. CoEP pays a one-off synthesis cost $C_{\text{build}}$ per assignment, then repairs the covered fraction at *zero* marginal LLM cost locally and pays a per-submission inference only on the abstaining tail.

Let $\Delta$ be the marginal per-submission saving of CoEP over a competitor and $f$ the abstention (fallback) rate; synthesis wins once the enrollment exceeds $N^* = C_{\text{build}}/(\Delta(1-f))$. On the Refactory corpus the abstention rate is $f \approx 2.2\%$ (local RR 97.75%). *[Table 2's LLM-cost column, per-assignment $f$, and per-assignment $N^*$ are populated from the full run.]* Synthesis is cheap: on the order of \$0.1 per assignment at these tiers, the price of a handful of baseline repairs, paid once *[exact per-model build costs from the full run]*. The structure dominates at judge scale. Priced at the standard \$3/\$15 per Mtok rate rather than the introductory one, our measured PyDex usage of 12,808 input and 5,798 output tokens per submission costs \$0.125 per submission, so sending Codeforces' 12.6M archived submissions through a per-query repairer costs about \$1.6M per pass, and the bill recurs with every cohort. The amortized route pays a one-off build per problem, on the order of \$2.5K for its 10,024 problems, plus a per-submission call only on the escalating tail. At the abstention rate we measure on the Refactory corpus that tail is another $\sim$\$30K, still two orders of magnitude below a full pass *[Codeforces abstention rate from the full run]*.

Beyond dollars, the covered fraction is repaired in about a second on a CPU with no network dependency, rate limits, or nondeterminism. A hosted per-query API offers none of these, and they matter most at deadline-night submission spikes.

**RQ2.** CoEP replaces a per-submission inference with a one-off cents-scale synthesis plus a fallback-only tail, breaking even at $N^* = C_{\text{build}}/(\Delta(1-f))$, past a handful of submissions per assignment at measured $f$ and build cost.

**RQ3: Can Caching and Cheaper Tiers Close the Gap?** The standard remedies for per-submission cost are provider prompt caching and cheaper model tiers, and our design measures both. $\text{PyDex}_{\text{PC}}$ re-prices every PyDex query under the provider's documented caching rules, as an upper bound that assumes a cache hit wherever one is allowed; every LLM arm already runs at the cheapest tier at which caching functions. *[$\text{PyDex}_{PC}$ vs. PyDex vs. CoEP cost cells from the full run.]* The measurement exposes why the discount is small. Caching pays for a shared leading *prefix*, but the shared material in these prompts is not a prefix: the variable buggy program and retrieved exemplars come first, leaving an identical prefix of about ten tokens. What remains cacheable is the reuse within one submission across its $K$ sampled candidates, and that is what $\text{PyDex}_{\text{PC}}$ realizes. The linear structure is untouched either way, since every submission still triggers a model round trip with its residual cost, latency, and data egress.

The break-even bound makes the same point analytically. $N^* = C_{\text{build}}/(\Delta(1-f))$ is invariant to token price under the stated assumption that build and per-query tokens are billed on the same schedule: $C_{\text{build}}$ and $\Delta$ scale together, so a cheaper model or provider caching shrinks both sides equally and does not change $N^*$. The claim is about how many model calls per specification are made, not the price of a call. Because every arm runs at the cheapest tier, the measured dollar gaps are a *floor* on the effect, not its size.

**RQ3.** No. Caching and cheaper tiers shrink the per-call constant, and $\text{PyDex}_{\text{PC}}$ realizes that discount, but every submission still buys a model call; the break-even $N^*$ is price-invariant, so the structural gap persists at any tier.

**RQ4: Component Contributions and Rule Analysis** **Two complementary local mechanisms.** CoEP's local repair combines named rules (problem-specific plus the shared generic library) with reference-guided transplant, and the two carry different loads. The rules make small, labelled edits for

the common misconceptions, such as a relaxed comparison or an off-by-one bound, at very small patch size (mean RPS $\approx$0.03). But most student bugs are not single-token slips; they need a medium, structural edit no token-level rule expresses, and there the reference-guided transplant supplies it by adopting the smallest set of differing hunks from a structurally-matched correct program, variable-renamed to fit. Across the whole benchmark the rule layer fixes 15.8% of repaired submissions (447 of 2,833) at mean RPS 0.063, and the transplant the remaining 84.2% at 0.570, a ninefold difference in how much of the student's program is disturbed. The transplant is what lifts a rule-only tool to competitive coverage; the rules keep the common cases at near-minimal patch size and carry the diagnosis below. The split is stable across builders: the cheaper model's rules fire on 13.8% of repairs at RPS 0.068, within two points of the frontier model's. A *framework-only* ablation, the identical tool with the LLM-authored rules removed, isolates the LLM layer: on this corpus it adds nothing to coverage and almost nothing to patch size (RR 97.75% vs. 97.75%; mean RPS 0.326 vs. 0.329), because the transplant absorbs the cases the rules handled, and the always-on generic library accounts for two-thirds of the rule-layer firings (143 of 220). What the build measurably buys here is therefore not coverage but diagnosis, the misconception-labeled repairs, plus the strategy-matched escalation path; whether LLM-authored rules buy coverage where correct pools are sparse is precisely what the harder Codeforces half of our benchmark tests (RQ5).

**Diagnosis for free.** Every rule-produced patch carries the ordered rule names that produced it, so aggregated over a class the tool yields a misconception profile: how often each named mistake occurred, and on which programs. An instructor gets this report at no extra cost, and a per-submission API never emits it. Diagnosis coverage, the fraction of local repairs carrying a misconception label, is about 13% on this corpus, concentrated on the most frequent slips. This is a property of the artifact, not the coverage number: it is why a local, auditable repairer is worth synthesizing even where a hosted model would pass the same tests.

**What is authored, what recurs.** Every repair event logs the ordered rule names that produced it, split by origin (builder-authored vs. generic), so we can report which misconceptions the builder anticipated, which rules recur across problems (the generic comparison and bound slips), and which fire most often per assignment. What the fallback had to handle instead, and what of it became a rule, is RQ6. *[Rule-provenance tables from the full run's event logs.]*

**Builder capability.** Synthesis is run with the same two models as every LLM arm (GPT-5.4-mini, Claude Sonnet 5). Showing the builder *worked (buggy $\rightarrow$ fixed) pairs* rather than raw buggy samples turns "infer the fix, then generalize" into "generalize this shown repair"; in earlier probes with the absolute cheapest tiers this lifted build-sample coverage from single digits to 75%, and the shared generic library plus the reference engine make the local path robust to builder weakness regardless. *[Per-builder build-sample coverage and rule quality from the full run.]* Capability is bought once per assignment, not once per submission; that is the entire point.

**RQ4.** Named rules produce 15.8% of local repairs at mean RPS 0.063, ninefold smaller than the transplant's 0.570 and each carrying a misconception label; reference-guided transplant produces the other 84.2% and is what carries coverage. The division is the same under both builders.

**RQ5: The Escalation Threshold** The local tool accepts whatever passes the tests, so its patch-size distribution has a tail the acceptance rule does nothing about: on refactory_q1 the largest local patch reaches RPS 2.00 and on refactory_q2 3.44, patches larger than the program they repair. The tail is thin but it moves the mean, which sits well above the median on every assignment (0.141 vs. 0.218 on refactory_q1). $\tau_p$ (Eq. 1) is what routes that tail to the fallback.

Calibration behaves as an outlier rule should: the threshold and the resulting rate are set by each assignment's own distribution rather than by a target. Across the eleven assignments $\tau_p$ ranges from 0.44 to 1.71 and fires on between 1.0% and 12.4% of repaired submissions, firing least where the tool's patches are already uniformly small (1.0% on refactory_q3, whose largest local patch is 0.975). On cf_10C, whose three submissions leave too little to estimate a median and a spread from, calibration declines to define a threshold at all and the gate stays off; that is the designed behaviour, not a special case. It is also close to a property of the assignment rather than of the builder: the two builders agree to the fourth decimal on refactory_q2 and refactory_q3, and to within 4% on three of the remaining four CS1 assignments.

**What escalation buys.** Over the full benchmark the gate fires on 206 submissions, 7.3% of those repaired locally, and the model returns the smaller patch on 135 of them (65.5%), for $1.31 of spend. Per assignment the effect is largest exactly where the local patches were worst: refactory_q4 moves from 0.408 to 0.287 and refactory_q5 from 0.196 to 0.158, while refactory_q3, which barely fires, moves 0.286 to 0.284.

One row needs reading carefully. On refactory_q2 the arm's mean RPS *rises*, 0.450 to 0.463, which looks like the gate doing harm. It is not. Among the 379 submissions both configurations repair, the gate enlarged none and pulled the mean down to 0.438; the rise comes from 25 submissions only the fallback repairs, whose patches average 0.828. $\text{CoEP}_{\text{LF}}$ is averaging over a harder set than $\text{CoEP}_{\text{L}}$ because it declines to leave those submissions unrepaired. Per submission the gate is monotone by construction; an arm's mean over a larger repaired set is not, and the two should not be read as the same quantity.

**Sensitivity.** Sweeping the outlier cutoff $k$ in $\text{median} + k\,\text{MAD}$ from 0.5 to 5 traces the frontier: $k$=0.5 reaches RPS 0.280 but escalates 38.0% of submissions, $k$=5 escalates 1.9% and recovers almost nothing, and the gain per point of escalation peaks at the conventional $k$=3 (0.0040 per point, against 0.0023 at $k$=2 and 0.0037 at $k$=4). That peak is an observation about these assignments, not the reason for the choice: $k$=3 is the standard cutoff and was fixed before the sweep was run. The conclusion is insensitive to it, since every cutoff in $[1, 4]$ leaves CoEP below both baselines' patch size at a fraction of their spend.

**RQ5.** A build-time outlier threshold routes 7.3% of locally repaired submissions to the fallback and gets a smaller patch back on 65.5% of them, for $1.31 across 2,872 submissions, without ever enlarging a patch or losing a repair. The cutoff is the conventional $3\,\mathrm{MAD}$, and the conclusion holds across $k \in [1, 4]$.

**RQ6: What a Term of Escalations Teaches the Tool** $\mathrm{CoEP}_{\mathrm{L+SL}}$ is the frozen tool re-run with the rules absorbed from its own fallback fixes. Because absorption happens outside the evaluated pass and the augmented tool is frozen before this arm runs, the comparison against $\mathrm{CoEP}_{\mathrm{L}}$ is order-independent: the same submissions, the same acceptance rule, two rule sets.

Only 3 of the 22 (assignment, builder) cells learned anything at all, and that is the first result: absorption needs a miss, and on nineteen cells the frozen tool leaves none. Where it does abstain, the rules transfer. On refactory_q2 three absorbed rules lift the frozen tool from 90.89% to 94.24% and pull mean RPS from 0.450 to 0.426; on cf_1011B a single rule, `binary_search_true_loop`, closes the last gap from 98.55% to 100.00%. The cheaper builder learns the same kind of thing more weakly: two rules on refactory_q2 for $+1.68$ points against the frontier model's $+3.35$.

Three submissions in total induced these rules, out of 486 evaluated across the three cells, so excluding them moves the rates by well under a point — the gain is measured almost entirely on programs the rules were never validated against. The learned names are also legible in the way the builder-authored ones are: `counter_var_name_confusion`, `uniqueness_off_by_zero`, `count_threshold_inverted`. They read as misconceptions, not as patches, which is what makes them reusable next term. Induction cost $0.63 in total across the three cells, priced to the artifact rather than to the arm it was gathered alongside.

**RQ6.** Absorption fires on only 3 of 22 cells, because elsewhere the frozen tool leaves no miss to learn from. Where it does fire, three rules take refactory_q2 from 90.89% to 94.24% and one takes cf_1011B to 100.00%, for $0.63; three inducing submissions out of 486 mean the gain is measured on programs the rules never saw.

**RQ7: Generalization and Robustness** The corpus is built to stress two axes the CS1 benchmark cannot. Difficulty: the six Codeforces problems climb from rating 800 to 2000, and the deep solution pools of the CS1 assignments give way to sparser, more diverse ones as difficulty rises. Provider: every LLM arm runs identically on GPT-5.4-mini and Claude Sonnet 5, so any conclusion that holds for one provider only would surface here. *[Per-problem RR/RPS by rating band and per-provider deltas from the full run; the classical symbolic baseline's per-band trajectory from the same pool.]*

One pattern is already visible in the completed arms. The symbolic baseline loses ground on Codeforces overall, repairing 78.6% against the 90.8% it reports on its own corpus, and its per-problem rate swings between 40.3% and 100% with no clean monotone trend in rating. What predicts the collapse is not difficulty itself but whether the correct pool holds a structurally matching reference, since block-level reuse starves when solution strategies diverge. CoEP's hunk-level transplant faces the same pool sparsity, which is precisely where the LLM-authored rules and the fallback tail must carry weight; the per-band decomposition quantifies how much of the advantage survives *[completed from the full run]*.

**RQ7.** *[Answer box populated from the full run: advantage retention across rating bands and agreement of conclusions across the two providers.]*

## Related Work

**APR for programming assignments.** The pre-LLM systems surveyed in the introduction (Yi et al. 2017; Gulwani, Radicek, and Zuleger 2018; Wang, Singh, and Su 2018; Hu et al. 2019; Ahmed et al. 2022) exploit per-assignment structure, and the reference-reuse ones among them depend on finding a structurally matching correct solution; our transplant stage stands openly in their reference-reuse lineage, at hunk rather than block granularity, and our head-to-head run shows the composed artifact exceeding the strongest of them on both coverage and patch size. Closest to our synthesis layer, Refazer learns per-assignment syntactic transformations from (buggy, fixed) student pairs with PROSE (Rolim et al. 2017); CoEP replaces the example-driven synthesizer with a cheapest-tier LLM that *names* the misconception behind each rule, and surrounds the learned layer with generic search, reference transplant, and harness-gated escalation. At AAAI the setting stays active from the per-submission side: counterexample-guided and preference-tuned minimal repair at AAAI-25 (Orvalho, Janota, and Manquinho 2025; Dai et al. 2025), and AAAI-26's learner-tailored repair, which retrieves per-assignment example fixes to guide each query (Dai et al. 2026), amortizing *data* into a retrieval index while every repair still pays for an LLM call (our PyDex arm bounds this design's economics). CoEP instead removes the per-submission call for everything its artifact covers and pays only on escalation.

**LLM-based repair.** AlphaRepair recast repair as generation (Xia and Zhang 2022); ChatRepair adds conversational test feedback and per-bug cost accounting (Xia and Zhang 2024); PyDex/MMAPR target student programs (Zhang et al. 2024, 2022); FastFixer accelerates per-query inference (Liu et al. 2024); CREF adds a tutor in the loop (Yang et al. 2024). All are per-submission; CoEP is complementary (any can serve as its fallback) but shifts the primary cost to the assignment.

**Templates, cost-efficient inference, and tool synthesis.** PAR and TBar showed a fixed library of human-curated fix templates repairs a surprising share of real bugs (Kim et al. 2013; Liu et al. 2019); CoEP's generic layer is a mutation-inversion instance (Just et al. 2014), and its novelty is the *synthesized, assignment-specific* template layer, authored per specification by an LLM. FrugalGPT cascades, SRepair's cheap pipelines, and provider caching (Chen, Zaharia, and Zou 2024; Xiang et al. 2024; Anthropic 2025; OpenAI 2024) all optimize *within* the pay-per-query paradigm and compose with CoEP, which instead removes queries. Voyager and

LLMs-as-tool-makers amortize agent-written code across episodes (Wang et al. 2024; Cai et al. 2024), self-refinement improving one-shot generations (Madaan et al. 2023; Shinn et al. 2023); CoEP adds what education forces: verifiability (never-trust-self-reports) and minimal, measured patches, because its output is feedback to a learner.

## Ethical Statement

CoEP automates feedback, not grading, under instructor oversight; unhandled students get full-LLM treatment; repairs are auditable; data are public.

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## Reproducibility Checklist

**General paper structure.** Conceptual outline of the approach: yes (Figure 1 and the Approach section's stage specification). Claims vs. speculation delineated: yes (RQ answer boxes state measured claims; the break-even analysis in RQ2–RQ3 is marked as derivation over measured quantities). Pedagogical references for background: yes (Related Work).

**Theoretical contributions.** This paper makes no formal theoretical claims; the break-even expression in RQ2 is elementary algebra over measured quantities, with all symbols defined in the text.

**Datasets.** All experiments use two publicly available corpora: the Refactory dataset (Hu et al. 2019) and Codeforces problems and submissions from the open-r1 release (Penedo et al. 2025). Our scripted import pipeline is released, covering test parsing, per-test entry points, and the oracle-fidelity audit, and we state the problem-selection rule we applied by hand, so the exact evaluation sets are reconstructible from the public corpora. No new human-subject data are collected.

**Computational experiments.** Source code, prompts, synthesized repairers, raw LLM responses, the ported Refactory-tool configuration, and analysis scripts are released; experiments run on CPU only. Model versions, pricing snapshot, tokenizer convention, decoding budget (single sample per call), execution timeouts, and the no-split whole-pool protocol are stated in Experimental Setup and fixed in the released configuration. Metrics are formally defined; repair rates carry 95% Wilson intervals and paired claims use exact McNemar tests. Local search and evaluation are deterministic. Builder synthesis is stochastic; every round's prompt, response, and build-sample outcome for both builder tiers is released with the artifacts.